

A population proportion

A population proportion p is the fraction of population members with a specified outcome.

What outcome would define p when studying use of a campus printing service?

A sample proportion

The sample proportion is the number of observed successes divided by the sample size.

$$\hat{p} = \frac{X}{n}$$

For 128 successes among 400 observations, what is the sample proportion?

Two notation choices

This textbook uses p' for the sample proportion; $\hat{p}$ denotes the same statistic.

$$p' = \hat{p}$$

Which symbol refers to the unknown population proportion?

Count or proportion?

Synthetic random sample: 128 of 400 students used the printing service this week.

Which number is a success count, and which number is the sample proportion?

A binary outcome

A proportion records whether each observation meets a specified condition.

Would average pages printed per student be a proportion problem?

Sampling variability of proportions

For independent binary observations with success probability p , the sample proportion has mean p and SD $\sqrt{p(1-p)/n}$.

$$SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

What happens to this SD when n quadruples?

An estimated standard error

A proportion interval estimates the standard error by replacing unknown p with the observed sample proportion.

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

For $\hat{p}=0.32$ and $n=400$, what is the estimated standard error?

Successes and failures

The textbook requires more than five observed successes and more than five observed failures for its ordinary normal-approximation proportion interval.

Does a synthetic sample with 128 successes and 272 failures pass this check?

A sparse success sample

Synthetic random sample: 2 successes and 38 failures.

Would a normal-approximation proportion interval meet the success-failure check?

Independence for proportions

Random sampling supports representativeness; independence also matters for the standard error formula.

Would surveying every member of a few randomly selected clubs automatically give independent responses?

A sample fraction

For sampling without replacement, $n \leq 10\%$ of the population is a common independence approximation.

Would a sample of 400 from 20,000 students meet that guideline?

A proportion margin of error

A normal-approximation proportion margin is a critical z value times estimated standard error.

$$EBP = z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

For $\hat{p}=0.32$, $n=400$, and $z^*=1.96$, what is EBP?

Constructing a proportion interval

Synthetic independent random sample: 128 of 400 students used printing this week. The population is large; use 95% confidence and $z^*=1.96$.

What is the normal-approximation confidence interval for p ?

Interpreting a proportion interval

Synthetic result: a 95% interval for printing-service use is 0.274 to 0.366.

How would you state this interval in percentages for the population?

Percentage points

Synthetic result: an estimated proportion is 32%, with margin of error 4.6 percentage points.

What are the percentage endpoints?

Confidence is about a procedure

A valid 95% proportion interval procedure captures the fixed p in about 95% of repeated intervals.

Does 95% mean that 95% of students use the printing service?

The plus-four adjustment

A plus-four interval uses two added successes and two added failures in its adjusted proportion.

$$\tilde{p} = \frac{x + 2}{n + 4}$$

For a synthetic sample with 8 successes among 40 observations, what is the adjusted proportion?

When to use plus four

The textbook recommends plus four when confidence is at least 90% and the sample size is at least 10.

Would a sample of 24 observations at 85% confidence meet these recommendations?

A plus-four interval

Synthetic independent random sample: 8 of 40 students used printing this week. Large population; 95% confidence; $z^*=1.96$.

$$\tilde{p} \pm z^* \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}}$$

What plus-four confidence interval for p follows?

Changing confidence

Use the same observed sample proportion and sample size. Compare 90% and 99% confidence intervals.

Which margin of error is larger?

Changing sample size

Hold the observed proportion and confidence level fixed. Increase n from 400 to 1,600.

How does the estimated margin of error change?

Planning a proportion sample

A planning estimate p_0 gives a sample-size formula, rounded upward, for a desired margin E .

$$n = \frac{(z^*)^2 p_0 (1 - p_0)}{E^2}$$

For $p_0=0.32$, $E=0.03$, and $z^*=1.96$, what n is needed?

Planning without an estimate

When no planning estimate is available, $p_0=0.5$ gives the largest value of $p_0(1-p_0)$.

Why would this choice be conservative for planning sample size?

A conservative planning problem

Plan a 95% proportion interval with margin at most 0.03. No prior proportion estimate is available; use $z^*=1.96$.

What sample size should be planned?

Precision cannot remove bias

Synthetic online poll: 90% of 5,000 voluntary respondents favor longer library hours.

Would its small calculated margin of error establish representativeness of all students?

A claim and an interval

Synthetic result: a 95% interval for printing-service use is 0.274 to 0.366. A claim states $p=0.50$.

Is the claimed proportion compatible with this interval?

Estimate versus null model

For a proportion interval, standard error uses $\hat{p}$. A proportion test may instead use a hypothesized p_0 .

Why could two analyses of the same sample use different standard errors?